Improved Model – Week 4

Group	Group 2 - AI Exam Answer Grader
Roll Numbers	J008, J009, J010, J011, J012
Class & Course	B.Tech DS 4th year – Speech & Natural Language Processing (SNLP)

1. Improved Approach

The Week 2 baseline (TF-IDF cosine similarity + Logistic Regression) and the Week 3 model (all-MiniLM-L6-v2 sentence-embedding cosine similarity + Logistic Regression) both reached about 50% test accuracy. Both give the classifier a single number per answer, so it cannot tell a short answer that hits one point from a complete answer.

The Week 4 model (V2) gives the classifier seven features per answer, implemented in src/features.py:

Feature	**What it measures**
MiniLM cosine	Semantic similarity of the whole answer to the reference (Week 3 feature)
TF-IDF cosine	Word-level similarity of the whole answer (Week 2 feature)
Key-point coverage	The reference is split into key points (lines/sentences); each is matched to the most similar student sentence using MiniLM, and the best matches are averaged
Word overlap	Share of reference words used by the student
Student length, reference length, length ratio	How complete the answer is relative to the reference

Key-point coverage implements Approach B from our Week 1 proposal: scoring against individual rubric concepts rather than overall similarity. Key points with similarity below 0.5 are returned as missing_key_points in the JSON feedback. The features are standardised and fed to a Logistic Regression. Its settings (C and class weighting) were chosen on validation macro-F1 (selected: C = 1.0, no class weighting). The test set was used once, for the final comparison. All three models are trained and evaluated by src/compare_models.py on the same EngSAF splits (4733 train / 338 validation / 506 test); the Baseline and V1 reproduce our Week 2 and Week 3 results exactly.

2. Baseline Model vs Improved Model Comparison

V2 improves on the baseline on all five metrics, on both the test and validation sets. Macro-F1 weights the three classes equally; QWK (quadratic weighted kappa) measures agreement with the human grade and penalises large errors more; MAE is the mean error in label points (lower is better); partial recall is the share of partially correct answers recognised.

Validation Set (338 answers)

Model	**Accuracy**	**Macro-F1**	**QWK**	**MAE**	**Partial Recall**
Baseline – TF-IDF + LR (Week 2)	0.5059	0.4820	0.4014	0.5888	0.1875
V1 – MiniLM + LR (Week 3)	0.5059	0.4966	0.4452	0.5533	0.2422
V2 – Hybrid features + LR (Week 4)	**0.5828**	**0.5766**	**0.4793**	**0.4941**	**0.3750**

Test Set (506 answers)

Model	**Accuracy**	**Macro-F1**	**QWK**	**MAE**	**Partial Recall**
Baseline – TF-IDF + LR (Week 2)	0.5020	0.4688	0.3194	0.6225	0.2408
V1 – MiniLM + LR (Week 3)	0.5000	0.4752	0.3326	0.5988	0.3141
V2 – Hybrid features + LR (Week 4)	**0.5257**	**0.5005**	**0.3643**	**0.5711**	**0.3770**

The largest gain is on partially correct answers: the baseline recognised 24% of them, V2 recognises 38%.

3. Confusion Matrices (Test Set)

V2 correctly grades 72 of 191 partially correct answers, up from 46 for the baseline. Rows are the true label, columns the predicted label (0 = incorrect, 1 = partially correct, 2 = correct). Baseline – TF-IDF + LR (Week 2)

Baseline – TF-IDF + LR (Week 2)

True \ Predicted	0 (Incorrect)	1 (Partially Correct)	2 (Correct)	Total
0 (Incorrect)	48	18	47	113
1 (Partially Correct)	26	46	119	191
2 (Correct)	16	26	160	202

V2 – Hybrid features + LR (Week 4)

True \ Predicted	0 (Incorrect)	1 (Partially Correct)	2 (Correct)	Total
0 (Incorrect)	44	34	35	113
1 (Partially Correct)	26	72	93	191
2 (Correct)	14	38	150	202

V2 also makes fewer of the most serious errors: incorrect answers graded as correct fell from 47 to 35. The trade-off is against V1: correct answers graded as incorrect rose from 4 to 14 (baseline: 16).

4. Ablation: Which Features Helped

The improvement comes mainly from the word overlap and length features; key-point coverage did not improve the score. Each row adds features to the one above, with the same classifier settings as V2 (test set).

Features Used	**Accuracy**	**Macro-F1**	**QWK**
MiniLM cosine only	0.4941	0.4695	0.3329
+ TF-IDF cosine	0.5000	0.4749	0.3286
+ Word overlap + Length features	0.5316	0.5088	0.3793
+ Key-point coverage (Full V2)	0.5257	0.5005	0.3643

The learned weights confirm this. For the "correct" class (standardised features), the largest weights are MiniLM cosine (+0.511) and word overlap (+0.439), while key-point coverage is close to zero (−0.058). We kept key-point coverage in V2. Removing it now would mean choosing the model using test results. It is also what produces the missing_key_points feedback, so its value is explainability rather than accuracy.

5. Example Outputs

src/grade.py grades one answer with all three models side by side. V2 keeps the Week 3 JSON fields and adds model and missing_key_points.

Example 1 – Correct answer in different words

Question: What does supervised learning use to train a model? (10 marks)

Reference: A supervised machine learning algorithm relies on labeled training data to learn a function that maps inputs to outputs.
Student: Supervised learning uses labeled data to train models to predict outcomes.

Model	Score	Confidence
Baseline (Week 2)	5/10	0.39
V1 (Week 3)	10/10	0.60
V2 (Week 4)	10/10	0.60

The baseline under-marks a correct paraphrase because it matches words, not meaning.

Example 2 – Partially correct answer
Question: Privileged operations (6 marks)
Reference: three key points – write to the CR3 register; write to the IDTR register; access a virtual address in the kernel range.
Student: Writing to the CR3 register to change the page directory.

All three models gave 6 / 6. Only V2 flagged the problem:

{
"question": "Q2 Privileged operations",
"score": 6, "maximum_score": 6,
"feedback": "The answer is graded correct overall, but these reference points were not found: write to the idtr register to change location of idt; access a virtual address in the kernel range.",
"confidence": 0.51,
"model": "v2_hybrid",
"missing_key_points": ["write to the idtr register to change location of idt", "access a virtual address in the kernel range"]
}

6. Analysis and Conclusion

V2 is a real but modest improvement. It beats the baseline on every metric on both splits, with the biggest gain on partially correct answers (24% → 38% recall).

Limitations:

- Small test set. With 506 test answers, a 2–3 point change is close to the noise level. The result is credible because all five metrics improve together, on both validation and test.
- Unseen questions. About 87% of test questions do not appear in the training data, so the models must generalise to new questions. This limits all three approaches.
- Missing points are detected but not penalised. As Example 2 shows, V2 finds missing key points but the classifier barely uses this feature, partly because EngSAF graders were often lenient with short answers that hit one key idea.